ASTANA IT UNIVERSITY

«Approved»
by the Dean
__________ Syzdykova Z.A.
«______» of __________ 2024

SYLLABUS
Academic Year 2024-2025

1. General Information	
Course title	Machine Learning Algorithms
Degree cycle (level)/ major	Bachelor / 6B06301 Cybersecurity
Year, term	3, 7
Number of credits	5
Language of delivery:	English
Prerequisites	Probability and statistics
Postrequisites	Deep learning in Cybersecurity
Lecturer(s)	Darkhan Zholtyaev, PhD in robotics engineering, Assistant Professor E-mail: darkhan.zholtayev@astanait.edu.kz Aigul Mimenbayeva, MSc, Lecturer E-mail: A.Mimenbayeva@astanait.edu.kz Kengeskanov Madiyar, MSc, Lecturer Astana IT University, Department of Computational and Data Science Expo, C1 block, 3rd floor, office C1.1.332
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	**The course examines** the basic machine learning algorithms, various approaches and technologies for data analysis, their qualities, features and impact in various fields of science and technology.
2. Course Goal(s)	**The course goal is** to acquire the theoretical and practical knowledge in the field of artificial intelligence in general and in particular in the creating of algorithms capable of learning
3. Course Objectives:	**Students should study the following concepts:** - artificial intelligence; - machine learning; - deep learning fundamentals; - data analysis; - machine learning categorizes; - machine learning workflow; - applications of machine learning; - supervised, unsupervised algorithms; - computer vision.
4. Skills & Competences	**After completing this course, students should demonstrate competency in the following skills:** - Demonstrate knowledge of basic forensic techniques and investigation of security incidents. - Have to ability to analyze data and use machine learning techniques to detect anomalies and predict threats.

5. Course Learning Outcomes:	**Having completed the course, the student will be able to:** As an outcome of mastering the course, students will be able to apply machine learning methods to visualize their data, build graphs, and present the results qualitatively.
6. Methods of Assessment	**The expected learning outcomes for the course will be assessed** through graded activities and ungraded activities. The graded activities include exams, homework, practice, SIS, and mid-terms, end-terms. The ungraded activities will be used to monitor your progress. A variety of these ungraded assessment techniques may be employed, including problems to be completed during class, direct questioning of students, answering questions in class and discussions during office hours.
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and/or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). **Basic Literature:** 1. Oliver Theobald. Machine Learning For Absolute Beginners: A Plain English Introduction (Second Edition), Independently Published, 2021. 2. Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022. 3. An introduction to Statistical Learning with Applications in Python, G. James, D. Witten, T. Hastie, R. Tibshirani, J. Taylor. 2023rd Edition. 4. S. Raschka. V. Mirjalili. Python Machine Learning. Released December 2019. 5. Paul Crickard. Data Engineering with Python. 2020. **Supplementary literature:** 1. Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, Mathematics for Machine Learning, Cambridge University Press, T2, 2020. 2. Sebastian Raschka and Vahid Mirjalili. Python Machine Learning: Machine Learning and Deep Learning with Python, scikit learn, and TensorFlow 2, Third Edition, Packt Publishing, 2020. 3. T. Hastie, R. Tibshirani, J. Friedman. The Elements of Statistical Learning. Second Edition. 2009. 4. Ch. Bishop. Pattern Recognition and Machine Learning. 2024. E-Book. 5. K. M. Murphy. Machine Learning: A probabilistic prespective. 2023. E-Book.
1. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
2. **Course policy**	**Course and University policies include:** **Attendance:** Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric.

	In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as "not graded". In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean's office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. **Class work**: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes. **Being late on class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. **Attestation I and II:** Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. **Home work / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

	Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.* **Final exam:** The final exam for course will be taken in the form of a computer test organized with the Learning Management System (moodle.astanait.edu.kz). **Laptops and mobile devices** can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. **Online lessons** can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). **Contacting the Course instructor**: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.

** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3	Chapter 2.1 Chapter 2.2 Chapter 2.3 Chapter 2.4 Chapter 2.5 Chapter 2 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
4	Chapter 3.1, Chapter 3.2, Chapter 3.3, Chapter 3 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
5	Chapter 2.8 Chapter 2.9 Chapter 2 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
6	Chapter 4.1 Chapter 4.2 Chapter 4 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
7	Chapter 4.3 Chapter 4.4 Chapter 4.5 Chapter 4.6 Chapter 4.7 Chapter 4 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
8	Chapter 1.9 Chapter 5.1 Chapter 5.2 Chapter 5 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
9	Chapter 5.3 Chapter 5.4 Chapter 5.5 Chapter 5 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises

10	Review of the course	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1st attestation	**Assignments**:** assignment 1, assignment 2, practice, **Mid Term**	**60** 20 20 20 **40**	100
2nd attestation	**Assignments**:** assignment 4 (project), practice, **End Term**	**60** 40 20 **40**	100
Final exam*	Test		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

** *The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.*

* *Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	**Introduction to Machine learning.** Machine learning categorizes. Machine learning pipeline. Applications of Machine learning. Python Essentials for MLOps.	2	3	0	1	9
2	**Machine Learning Workflow**. Data acquisition, pre-processing, model training, evaluation, and deployment. Data acquisition and manipulation with Python.	2	3	0	1	9
3	**Supervised Learning.** Regression algorithms: Linear Regression, Logistic regression. Introduction to Probability & Statistics. Probability distributions, mean, variance, and standard deviation. Implementation of regression models using Python.	2	3	0	1	9
4	**Classification algorithms.** Decision trees. K-Nearest Neighbors (KNN) algorithm. Support Vector Machine (SVM) Algorithm. Concepts, strengths & weaknesses. Implementation of classification models using Python.	2	3	0	1	9
5	**Ensemble learning.** Types of Ensemble Methods. Random Forests. Gradient Boosting. Implementation of Ensemble Methods using Python. Model selection & Hyperparameter tuning techniques. Model Evaluation Metrics.	2	3	0	1	9
Midterm						

6	**Unsupervised Learning.** Types of Unsupervised Learning. Unsupervised learning applications. K-Means Clustering. DBSCAN (Density based clustering) in ML. Implementation of unsupervised algorithms using machine learning in Python.	2	3	0	1	9
7	**Dimensionality reduction techniques in machine learning.** PCA (Principal Component Analysis). t-SNE (t-Distributed Stochastic Neighbor Embedding).	2	3	0	1	9
8	**Deep Learning Fundamentals.** Introduction to Artificial Neural Networks: Structure, activation functions, and basic training concepts. Feedforward Networks and Backpropagation.	2	3	0	1	9
9	**Convolutional Neural Networks.** Architecture and applications for image recognition. Transfer Learning Implementations in Keras.	2	3	0	1	9
10	**General overview of Deep learning algorithms.** Computer vision, Natural Language Processing, Deep reinforcement learning.	2	3	0	1	9
	Endterm					
	Total hours: 150	**20**	**30**	**0**	**10**	**90**

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	**2**	**3**	**4**	**5**
1	Chapter 1.1, Chapter 1.2, Chapter 1.3, Chapter 1.4, Chapter 1 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
2	Chapter 1.5, Chapter 1.7, Chapter 1 - SUPPLEMENTARY EXERCISES	9	Andreas C. Müller, Sarah Guido. Introduction to Machine Learning with Python, 2022.	Exercises
3	Chapter 2.1 Chapter 2.2	9	Andreas C. Müller, Sarah Guido. Introduction to	Exercises

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **The final exam** for course will be taken in the form of a computer test organized with the Learning Management System (moodle.astanait.edu.kz).

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Darkhan Zholtyaev	Assistant Professor	26.09.2024	
Aigul Mimenbayeva	Lecturer	26.09.2024	
Madiar Kengeskanov	Lecturer	26.09.2024	

Head of the Department of Computational and Data Science Sergaziyev M.Zh.